

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Forward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_original_best_Fw	rcim_original	Fw	0.002769	0.002951	6.250
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
track1_best_Fw	rcim_track1	Fw	0.003014	0.003204	6.819

Forward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_fw	wave1_current_registry	Fw	0.003404	0.003855	7.551
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.003254	0.003553	7.232
residual_harmonic_mlp_fw	wave1_current_registry	Fw	0.003273	0.003563	7.266
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
periodic_mlp_harmonic_fw	wave1_periodic_mlp_harmonic_campaign	Fw	0.003254	0.003553	7.232

Backward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
track1_best_Bw	rcim_track1	Bw	0.005027	0.005212	11.860

Backward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_bw	wave1_current_registry	Bw	0.003586	0.004023	7.832
harmonic_regression_bw	wave1_current_registry	Bw	0.003678	0.004012	8.058
periodic_mlp_bw	wave1_current_registry	Bw	0.003574	0.004006	7.807
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.003536	0.003874	7.728
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
periodic_mlp_harmonic_bw	wave1_periodic_mlp_harmonic_campaign	Bw	0.003583	0.003925	7.875

Forward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003603	0.004031	8.028
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003330	0.003762	7.378
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003366	0.003800	7.450
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003335	0.003708	7.404
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003186	0.003438	7.077
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003266	0.003550	7.258

Backward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003742	0.004166	8.184
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.003626	0.004082	7.907
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.003555	0.003985	7.767
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003628	0.003987	7.979
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002392	0.002639	5.466
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002625	0.002877	6.013

Forward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003194	0.003499	7.083
residual_harmonic_gru_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.006983	0.009275	15.722
residual_harmonic_gru_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007869	0.010574	17.740
residual_harmonic_lstm_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003229	0.003533	7.164
residual_harmonic_lstm_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007042	0.009370	15.868
residual_harmonic_lstm_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007731	0.010235	17.430

Backward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003502	0.003857	7.654
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.008984	0.012987	20.358
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.009370	0.013165	21.267
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003440	0.003793	7.510
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.007367	0.009945	16.660
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010268	0.014769	23.355

Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
track1_best_Fw	rcim_track1	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.003254	0.003553	7.232

Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
track1_best_Bw	rcim_track1	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.003536	0.003874	7.728
periodic_mlp_bw	wave1_current_registry	Bw	0.003574	0.004006	7.807

Forward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
track1_best_Fw	rcim_track1	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003603	0.004031	8.028
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003330	0.003762	7.378
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003366	0.003800	7.450
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.003335	0.003708	7.404
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.003186	0.003438	7.077
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.003266	0.003550	7.258

Backward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
track1_best_Bw	rcim_track1	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003742	0.004166	8.184
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.003626	0.004082	7.907
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.003555	0.003985	7.767
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.003628	0.003987	7.979
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.002392	0.002639	5.466
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.002625	0.002877	6.013

Forward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
track1_best_Fw	rcim_track1	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.003053	0.003395	6.731
residual_harmonic_gru_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003194	0.003499	7.083
residual_harmonic_gru_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.006983	0.009275	15.722
residual_harmonic_gru_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007869	0.010574	17.740
residual_harmonic_lstm_sequence_sparse_rcim_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003229	0.003533	7.164
residual_harmonic_lstm_sequence_dense240_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007042	0.009370	15.868
residual_harmonic_lstm_sequence_dense360_Fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007731	0.010235	17.430

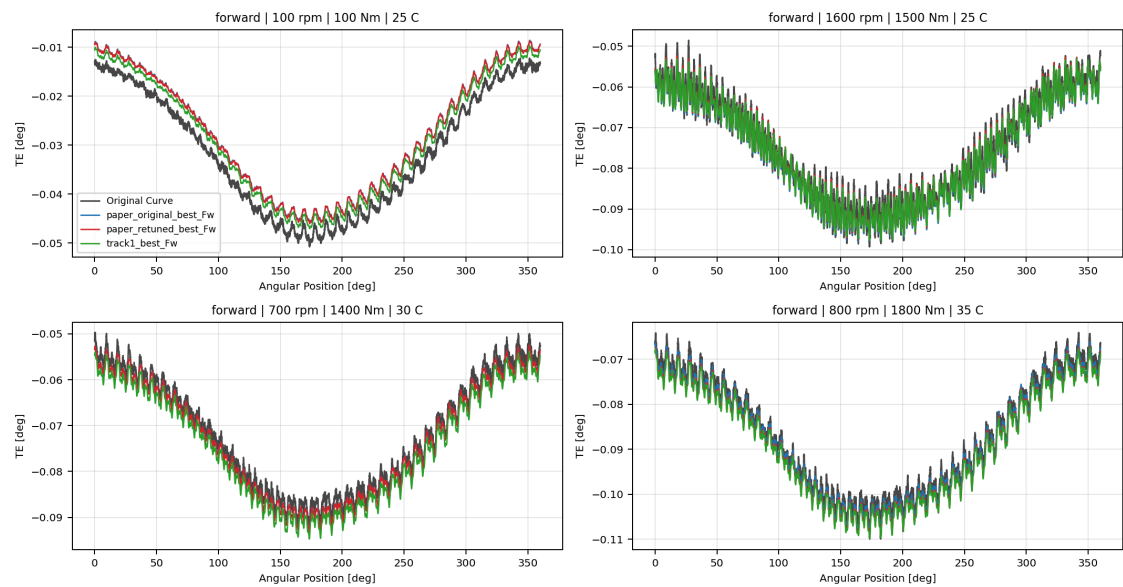
Backward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
track1_best_Bw	rcim_track1	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.003258	0.003651	7.051
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003502	0.003857	7.654
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.008984	0.012987	20.358
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.009370	0.013165	21.267
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003440	0.003793	7.510
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.007367	0.009945	16.660
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010268	0.014769	23.355

Comparison Gallery - Forward Reference Model Overlay

Included models: `paper_original_best_Fw` , `paper_retuned_best_Fw` , `track1_best_Fw` .

Forward Reference Model Overlay

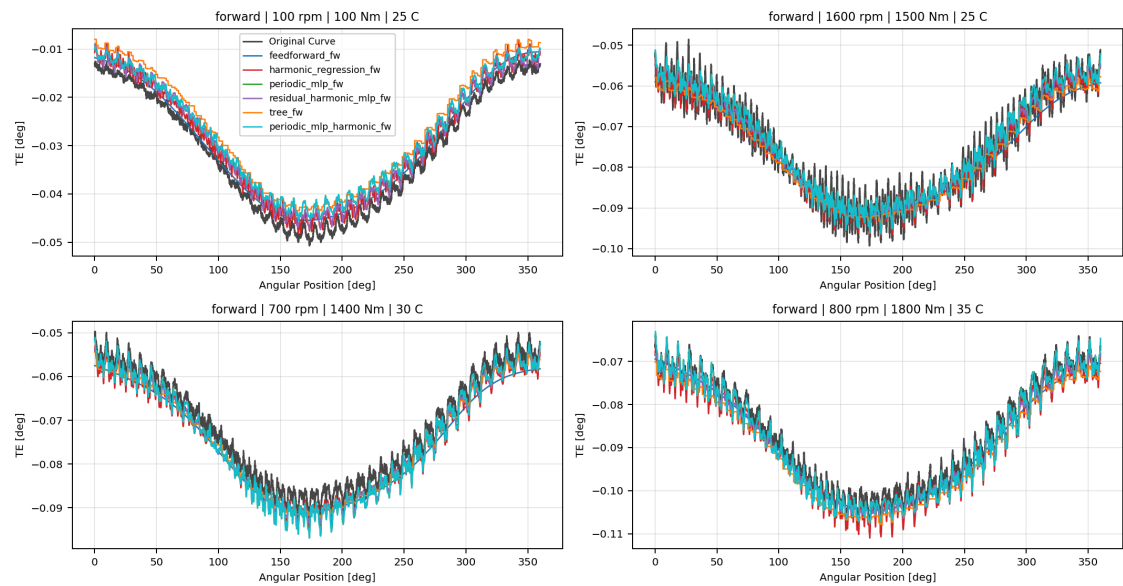


Forward Reference Model Overlay curve-verification comparison

Comparison Gallery - Forward Wave 1 Family Model Overlay

Included models: `feedforward_fw` , `harmonic_regression_fw` , `periodic_mlp_fw` , `residual_harmonic_mlp_fw` , `tree_fw` , `periodic_mlp_harmonic_fw` .

Forward Wave 1 Family Model Overlay

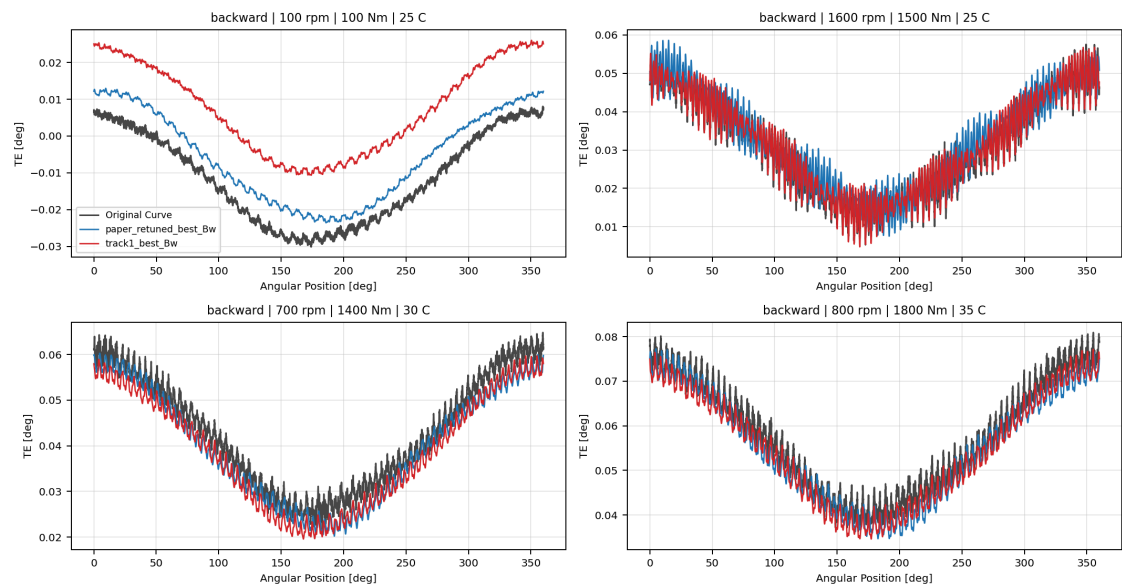


Forward Wave 1 Family Model Overlay curve-verification comparison

Comparison Gallery - Backward Reference Model Overlay

Included models: `paper_retuned_best_Bw` , `track1_best_Bw` .

Backward Reference Model Overlay

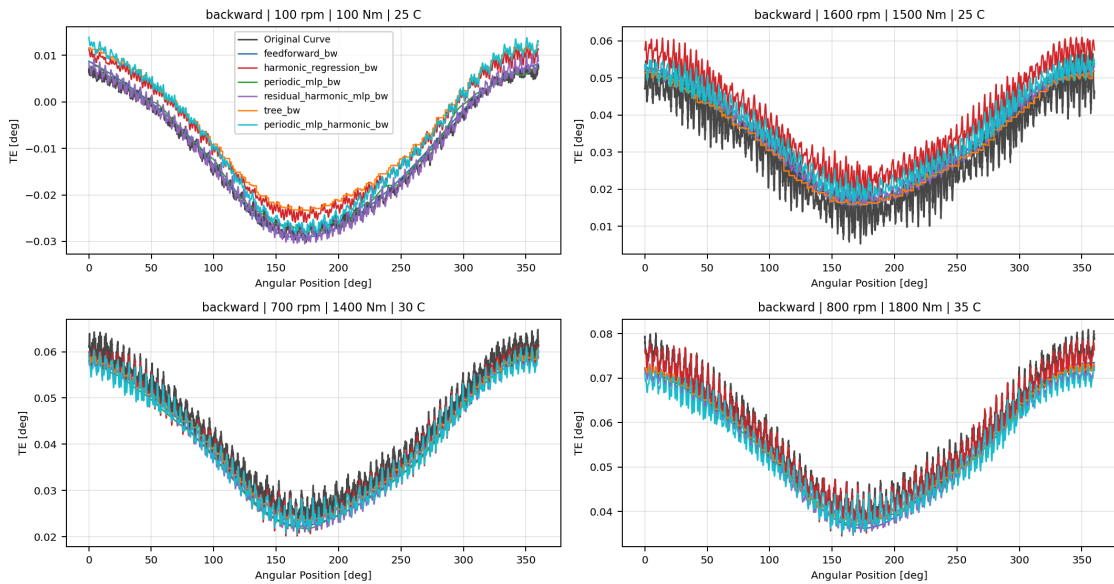


Backward Reference Model Overlay curve-verification comparison

Comparison Gallery - Backward Wave 1 Family Model Overlay

Included models: `feedforward_bw` , `harmonic_regression_bw` , `periodic_mlp_bw` , `residual_harmonic_mlp_bw` , `tree_bw` , `periodic_mlp_harmonic_bw` .

Backward Wave 1 Family Model Overlay

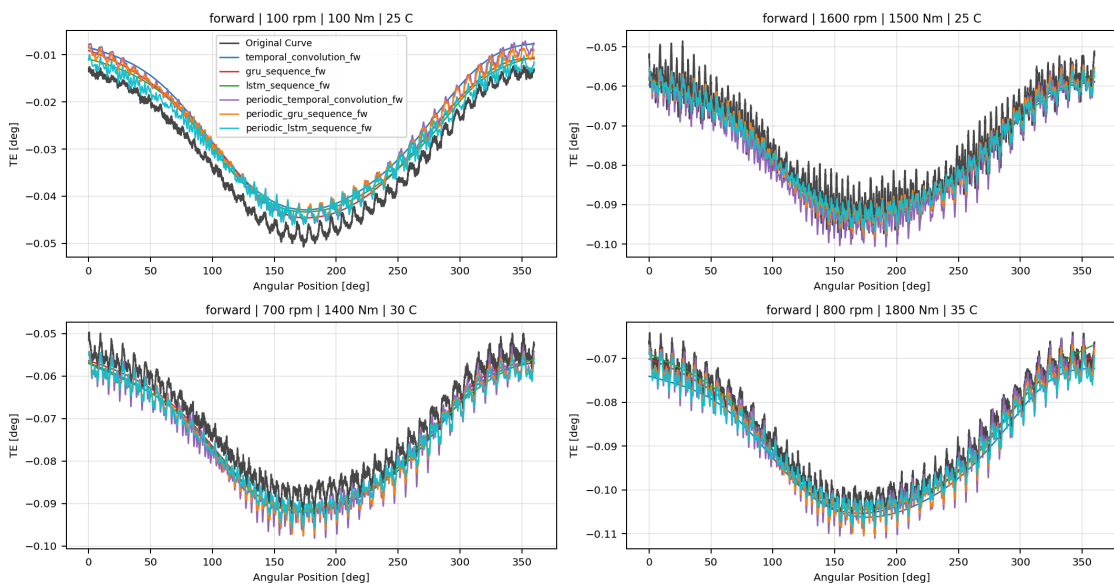


Backward Wave 1 Family Model Overlay curve-verification comparison

Comparison Gallery - Forward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_fw` , `gru_sequence_fw` , `lstm_sequence_fw` , `periodic_temporal_convolution_fw` , `periodic_gru_sequence_fw` , `periodic_lstm_sequence_fw` .

Forward Wave 2 Temporal Model Overlay

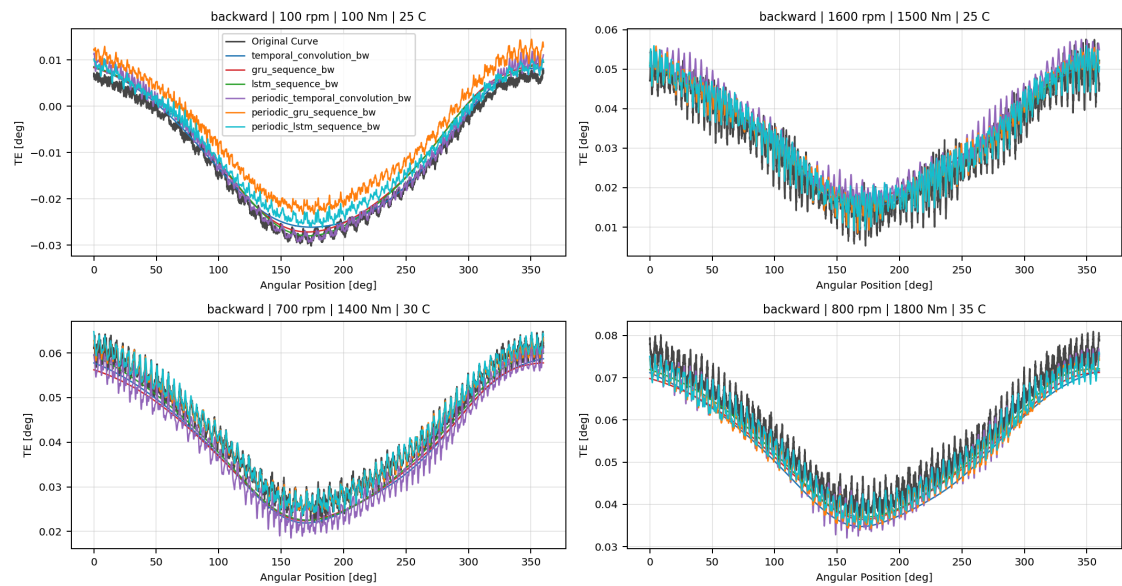


Forward Wave 2.1 Temporal Model Overlay curve-verification comparison

Comparison Gallery - Backward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Wave 2 Temporal Model Overlay

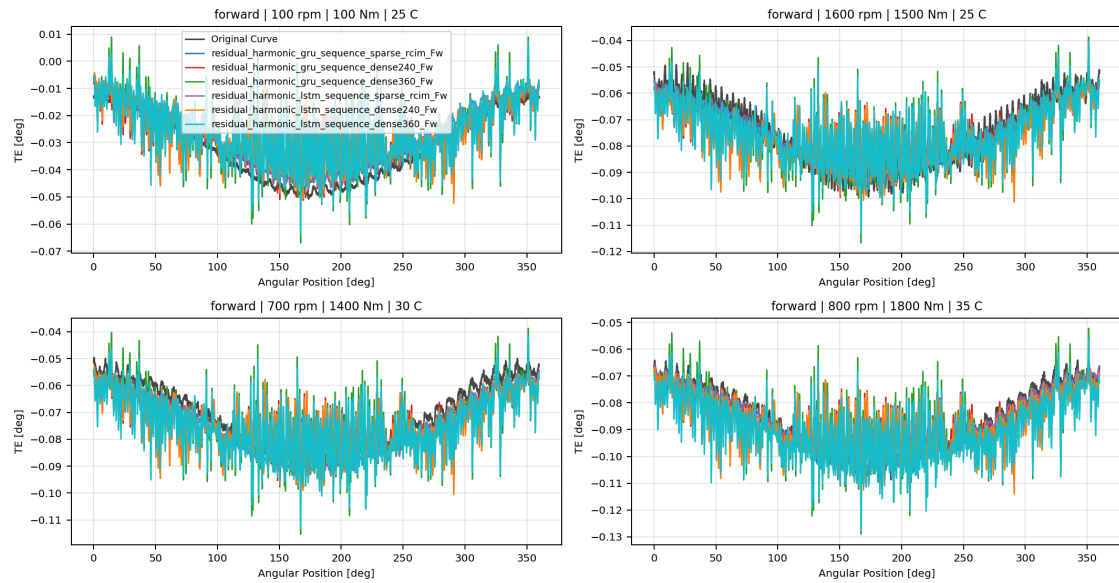


Backward Wave 2.1 Temporal Model Overlay curve-verification comparison

Comparison Gallery - Forward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Fw` ,
`residual_harmonic_gru_sequence_dense240_Fw` , `residual_harmonic_gru_sequence_dense360_Fw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Fw` , `residual_harmonic_lstm_sequence_dense240_Fw` ,
`residual_harmonic_lstm_sequence_dense360_Fw` .

Forward Wave 2C Residual Harmonic Temporal Overlay

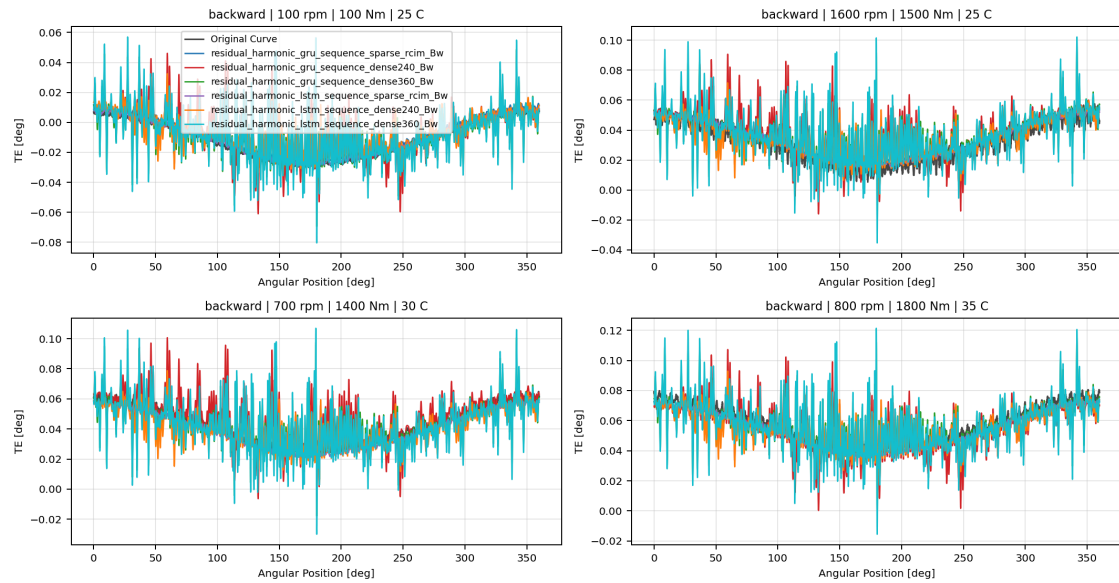


Forward Wave 2.3 Residual Harmonic Temporal Overlay curve-verification comparison

Comparison Gallery - Backward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Bw` ,
`residual_harmonic_gru_sequence_dense240_Bw` , `residual_harmonic_gru_sequence_dense360_Bw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Bw` , `residual_harmonic_lstm_sequence_dense240_Bw` ,
`residual_harmonic_lstm_sequence_dense360_Bw` .

Backward Wave 2C Residual Harmonic Temporal Overlay

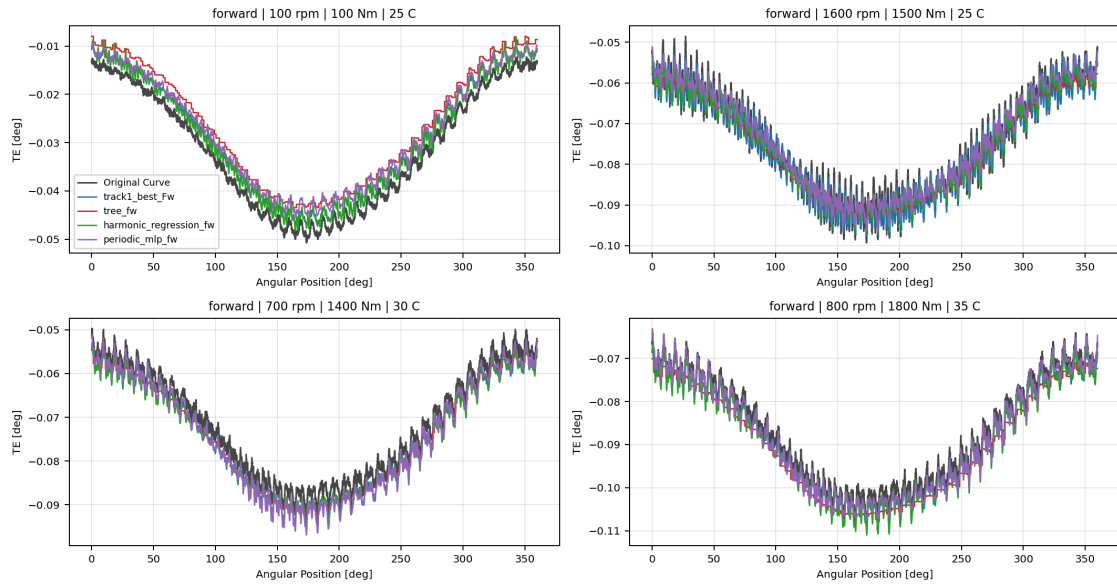


Backward Wave 2.3 Residual Harmonic Temporal Overlay curve-verification comparison

Comparison Gallery - Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: `track1_best_fw` , `tree_fw` , `harmonic_regression_fw` , `periodic_mlp_fw` .

Forward Track 1 And Screened Wave 1 Overlay

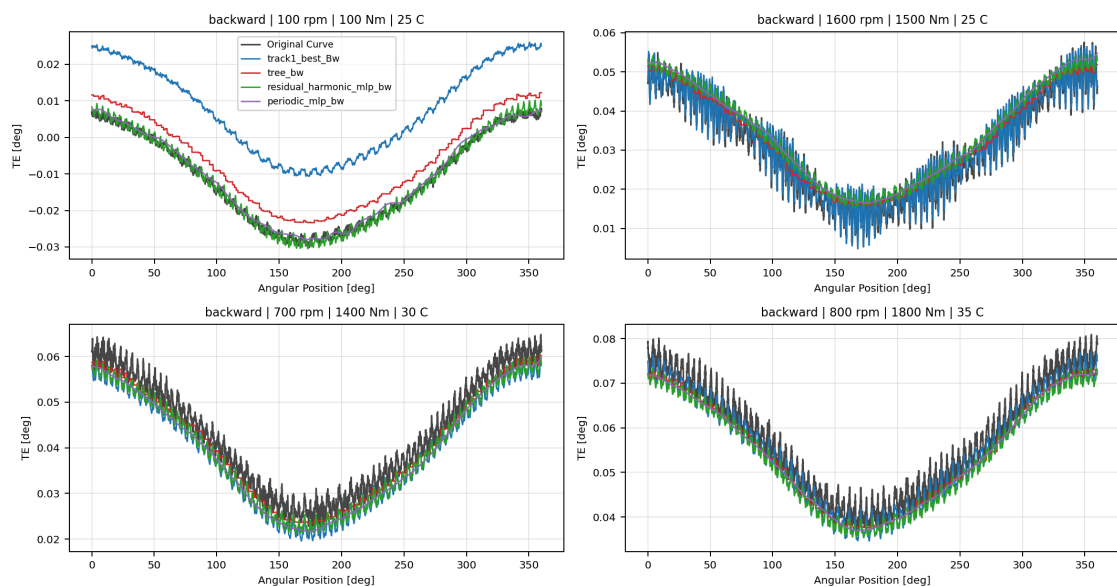


Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay curve-verification comparison

Comparison Gallery - Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: `track1_best_bw` , `tree_bw` , `residual_harmonic_mlp_bw` , `periodic_mlp_bw` .

Backward Track 1 And Screened Wave 1 Overlay

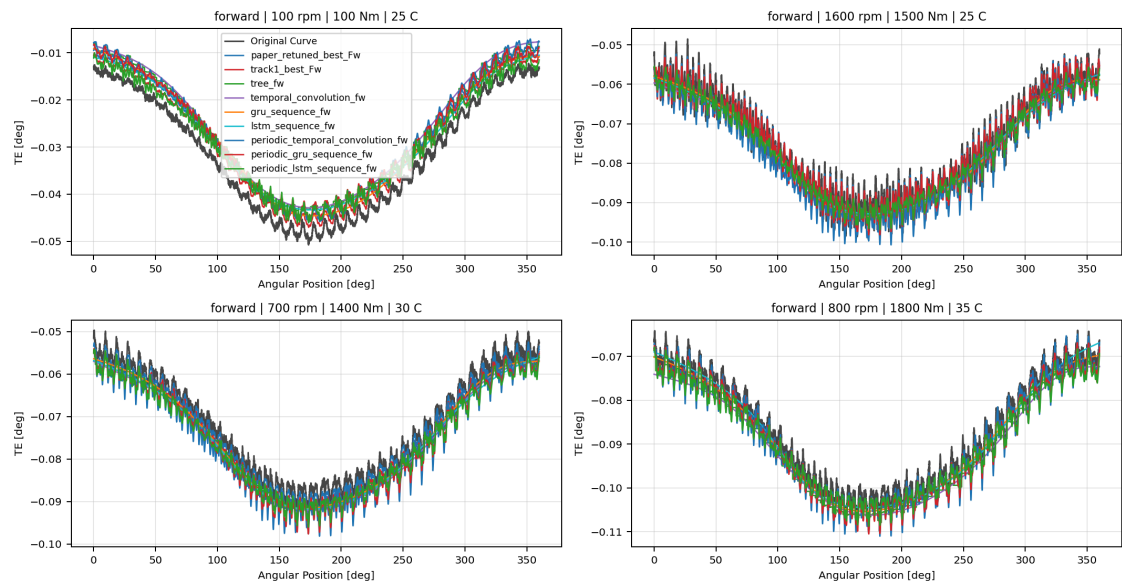


Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_Fw` , `track1_best_Fw` , `tree_fw` , `temporal_convolution_fw` , `gru_sequence_fw` , `lstm_sequence_fw` , `periodic_temporal_convolution_fw` , `periodic_gru_sequence_fw` , `periodic_lstm_sequence_fw` .

Forward Reference Tree And Wave 2 Overlay

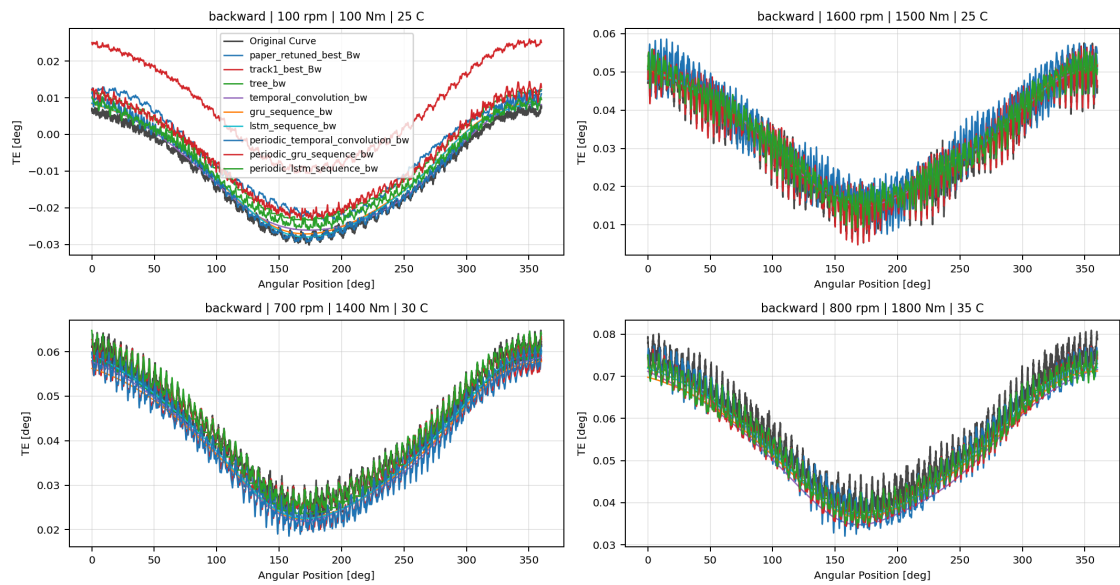


Forward Reference Tree And Wave 2.1 Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_Bw` , `track1_best_Bw` , `tree_bw` , `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Reference Tree And Wave 2 Overlay

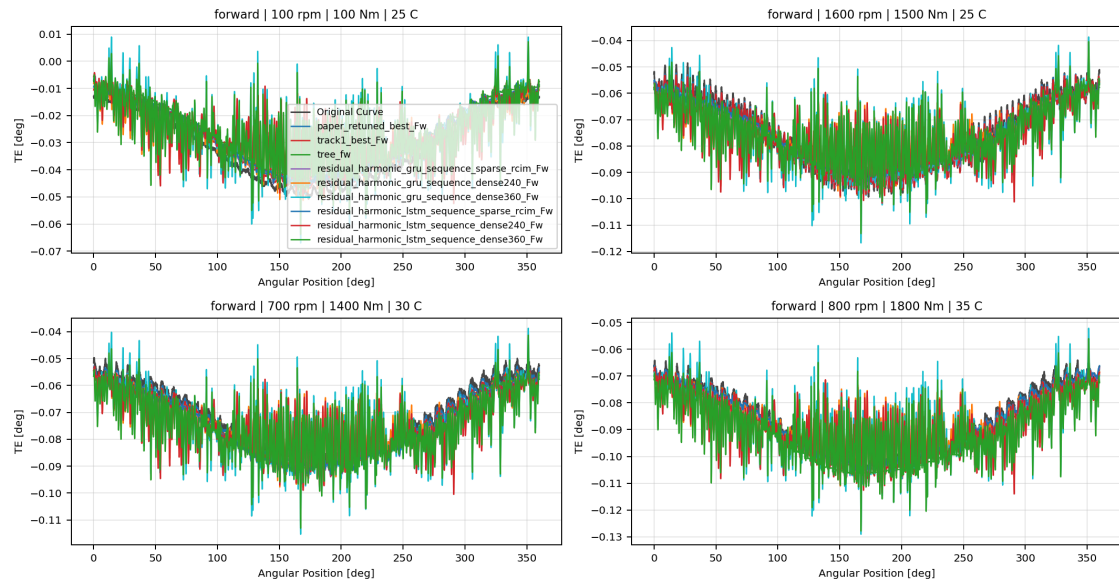


Backward Reference Tree And Wave 2.1 Overlay curve-verification comparison

Comparison Gallery - Forward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Fw`, `track1_best_Fw`, `tree_fw`,
`residual_harmonic_gru_sequence_sparse_rcim_Fw`, `residual_harmonic_gru_sequence_dense240_Fw`,
`residual_harmonic_gru_sequence_dense360_Fw`, `residual_harmonic_lstm_sequence_sparse_rcim_Fw`,
`residual_harmonic_lstm_sequence_dense240_Fw`, `residual_harmonic_lstm_sequence_dense360_Fw`.

Forward Reference Tree And Wave 2C Overlay

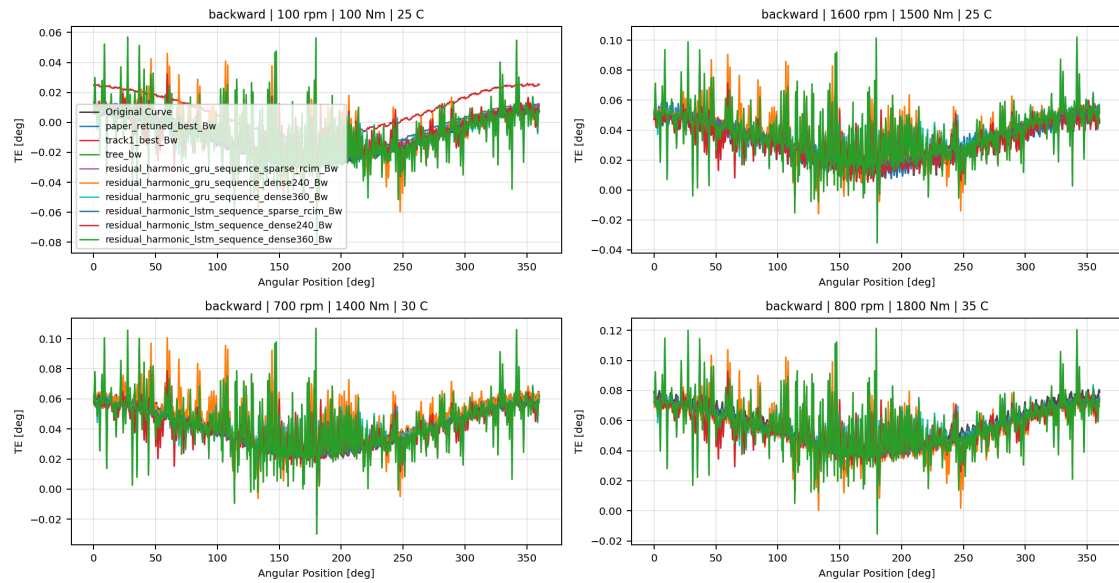


Forward Reference Tree And Wave 2.3 Overlay curve-verification comparison

Comparison Gallery - Backward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Bw`, `track1_best_Bw`, `tree_bw`,
`residual_harmonic_gru_sequence_sparse_rcim_Bw`, `residual_harmonic_gru_sequence_dense240_Bw`,
`residual_harmonic_gru_sequence_dense360_Bw`, `residual_harmonic_lstm_sequence_sparse_rcim_Bw`,
`residual_harmonic_lstm_sequence_dense240_Bw`, `residual_harmonic_lstm_sequence_dense360_Bw`.

Backward Reference Tree And Wave 2C Overlay



Backward Reference Tree And Wave 2.3 Overlay curve-verification comparison

Output Artifacts

- output directory: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-05-28-14-34-44__track2_multi_model_curve_comparison_report` ;
- summary YAML: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-05-28-14-34-44__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-05-28-14-34-44__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\multi_model_curve_comparison_report\[2026-05-28]\track2_multi_model_curve_comparison_report.md` .